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Least wall, most room: why bees build hexagons, and the honeycomb theorem

The free guide to the reel on honeycomb: how a hexagon grid compares with triangles and squares, who proved the honeycomb theorem, where architects use it, and a paper experiment you can do in ten minutes.



The question bees answer

A honeycomb divides a flat surface into cells of equal size. Wax is costly for the bees to make, so the less wall a cell needs, the more honey fits for the same wax.

The honeycomb theorem says a regular hexagonal grid has the least total perimeter of any way of dividing the plane into regions of equal area. Circles cannot tile without gaps, so they are not even a candidate.

The numbers

For cells of area 1, every wall is shared by two cells, so wall per cell is half the perimeter. Triangles need about 2.28, squares need 2.00, and hexagons need about 1.86 (the number 12 to the power 1/4).

Hexagons therefore use about 7% less wall than squares for the same area.

Who proved it

The idea is very old: Marcus Terentius Varro wrote about it in 36 BC and it is often credited to Pappus of Alexandria. In the 17th century Jan Brozek used a similar theorem to explain why bees build hexagons. Laszlo Fejes Toth proved a special case (convex cells) in 1943.

The full theorem was proved by Thomas C. Hales in 1999.

Built examples

Honeycomb in a hive	Cells of equal size tile the comb with the least wax wall.
Circle packing	The densest packing of circles also has each circle touching six others, which is why hexagon grids appear around it.
Hexagonal grids in design	Parametric facades and tilings use hexagon grids when they need equal-area panels with short joints.

Try it yourself: measure the wall

1. Draw a grid of 12 squares on graph paper, each 2 cm by 2 cm (area 4 cm2).
2. Draw 12 regular hexagons of the same area (side about 1.24 cm) on a second sheet, packed edge to edge.
3. Trace the lines with a coloured pen and measure the total length of line used, counting each shared wall once.
4. Divide the total by 12 to get the wall length per cell and compare: the hexagons should need about 7% less.
5. Try triangles of the same area and see which of the three needs the most wall.

The rule to remember

When you divide a space into equal cells, the shape of the cell decides how much wall you pay for.

Default answer	Use squares because they are easy to draw.
Form-finder answer	Ask which cell shape has the least boundary for the area you need.
Check	Ask: how much wall, panel joint or edge does my grid add per unit of area?

Go further

Honeycomb conjecture (Wikipedia)
https://en.wikipedia.org/wiki/Honeycomb_conjecture

Honeycomb (Wikipedia)
https://en.wikipedia.org/wiki/Honeycomb

Thomas Hales (Wikipedia)
https://en.wikipedia.org/wiki/Thomas_Callister_Hales

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Facts: Wikipedia (Honeycomb conjecture); wall-per-cell values computed from the geometry of regular tilings.

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